

Scientific Report: Phase 2 Corrosion Prediction Pipeline					
This report documents the complete Phase 2 deep-learning workflow for corrosion prediction, including rationale, quantitative results, interpretation, alternative solutions, and practical recommendations.					
1. Project Context and Objectives					
The objective is to predict corrosion progression and quantify corrosion severity for different cementitious materials/specimens observed over time. The data include weekly images and synchronized tabular measurements. The Phase 2 objective was to deploy a deeper scientific solution based on multimodal machine learning and deliver explainable outputs suitable for research communication.					
2. Data Summary and Material Definitions					
- Total observations: 792 - Total materials/specimens: 48 - Observed week range: 0 to 36 - Corrupted images detected and handled: 1					
Table 1 reports the full material names with complete treatment description, mesh configuration, and saline exposure information.					
Specimen	Material_Full_Name	Weeks_Observed	Week_Min	Week_Max	Coverage
L1	D01 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	7.1600
L2	D02 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	5.6100
L3	D03 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	5.9000
L4	D04 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	2.2800
L5	D05 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	4.2700
L6	D06 - No treatment (control) (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.5800
L7	E01 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.2200
L8	E02 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	4.3500
L9	E03 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	3.5600
L10	E04 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.1300
L11	E05 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	4.4900
L12	E06 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	9.3700
L13	F01 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	7.3100
L14	F02 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	7.9900
L15	F03 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.9500
L16	F04 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	5.7600
L17	F05 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.4100
L18	F06 - Surface-applied corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.3700
L19	G01 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	5.9400

Specimen	Material_Full_Name	Weeks_Observed	Week_Min	Week_Max	Coverage
G02	G02 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.6600
G03	G03 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	6.8300
G04	G04 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	4.5800
G05	G05 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	5.2300
G06	G06 - Painted protective treatment (steel meshes: 4, NaCl: 5.0%)	18	0	36	4.9400
S1MI01	S1MI01 - No treatment (control) (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.7400
S1MI02	S1MI02 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.5400
S1MI03	S1MI03 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.8000
S1MI04	S1MI04 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	1.6300
S1MI05	S1MI05 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.7100
S1MI06	S1MI06 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.3400
S1MI07	S1MI07 - Mixed-in corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	5.3400
S2SA01	S2SA01 - No treatment (control) (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.4200
S2SA02	S2SA02 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.3900
S2SA03	S2SA03 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	2.0200
S2SA04	S2SA04 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	5.2400
S2SA05	S2SA05 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.1100
S2SA06	S2SA06 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	2.8700
S2SA07	S2SA07 - Surface-applied corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.3000
S3PA01	S3PA01 - Painted protective treatment (steel meshes: 7, NaCl: 3.5%)	15	0	28	2.4600
S3PA02	S3PA02 - Painting + surface-applied inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.6000
S3PA03	S3PA03 - Painting + surface-applied inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.1600
S3PA04	S3PA04 - Painting + surface-applied inhibitor (steel meshes: 7, NaCl: 3.5%)	15	0	28	2.5100
S4SAVF01	S4SAVF01 - No treatment (control) (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.7000
S4SAVF02	S4SAVF02 - Surface-applied inhibitor + glass-based compound (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.9100
S4SAVF03	S4SAVF03 - Surface-applied inhibitor + glass-based compound (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.5500
S5VF01	S5VF01 - No treatment (control) (steel meshes: 7, NaCl: 3.5%)	15	0	28	3.4600
S5VF02	S5VF02 - Glass-based compound treatment (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.5900
S5VF03	S5VF03 - Glass-based compound treatment (steel meshes: 7, NaCl: 3.5%)	15	0	28	4.3800

3. Methodology and Scientific Rationale

3.1 Why this approach was selected:

- The dataset size is moderate (792 observations), which supports transfer learning instead of full CNN training from scratch.
- Corrosion signals are strongly visual (rust distribution and texture), therefore image representation is fundamental.
- Material and environmental context (treatment, NaCl%, steel mesh count, week) carry additional mechanistic information.
- Grouped splitting by specimen prevents data leakage across timepoints of the same material, producing realistic generalization estimates.
- The resulting pipeline supports reproducibility and deployment in a laboratory workflow.

3.2 Phase 2 modeling workflow:

- Extract 512-dimensional visual embeddings from a pretrained ResNet-18 backbone.
- Build tabular branch with scaled numerical and one-hot categorical variables.
- Train and compare two deep regressors: image-only baseline and multimodal fusion.
- Optimize on validation MAE with early stopping.
- Evaluate on test groups using MAE, RMSE, and R2.
- Generate per-material progression diagnostics and uncertainty-oriented residual analysis.

3.3 Candidate solutions considered:

- Solution A: Image-only deep learning (chosen best in this run).
- Solution B: Multimodal fusion (image + tabular deep model).
- Solution C: Gradient-boosted tabular baseline from Phase 1 for progression reference.
- Solution D: Future sequence models (temporal transformer/LSTM) for direct time-series forecasting.
- Solution E: Survival analysis for time-to-event prediction under censoring.

4. Quantitative Results

Table 2. Model comparison on held-out test specimens.

model	test_mae	test_rmse	test_r2
image_only_mlp	5.6255	9.2639	0.5237
multimodal_mlp	5.7476	9.8652	0.4599

Best model selected for deployment: **image_only_mlp** (MAE=5.6255, RMSE=9.2639, R2=0.5237).

Table 3. Performance by treatment group (test split).

Treatment	n	mae	rmse	mean_true	mean_pred
SA_VF	15	3.6217	5.0698	3.2797	5.0076
SA_PA	15	4.0014	6.7644	7.5041	4.5745
MI	45	4.3392	7.1858	8.1498	10.0041
SA	33	6.0240	11.4219	6.8120	7.1797

Treatment	n	mae	rmse	mean_true	mean_pred
PA	18	6.4422	8.1051	12.5159	7.2082
NO	33	8.1846	12.0128	16.2200	9.0896

Table 4. Performance by material series (test split).

series	n	mae	rmse	mean_true	mean_pred
S	105	4.0751	8.0377	6.9608	6.8779
F	18	5.5848	7.3224	4.3597	9.8911
G	18	6.4422	8.1051	12.5159	7.2082
D	18	13.8934	16.1835	26.6223	12.8102

5. Visual Diagnostics

Figure 1. Learning curves for both deep models.

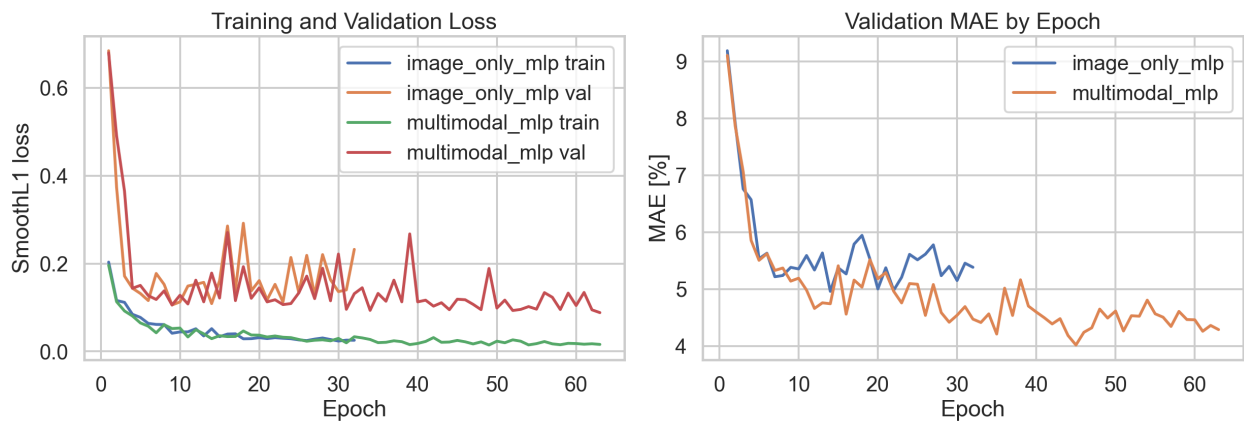


Figure 2. Predicted vs true corrosion on test specimens.

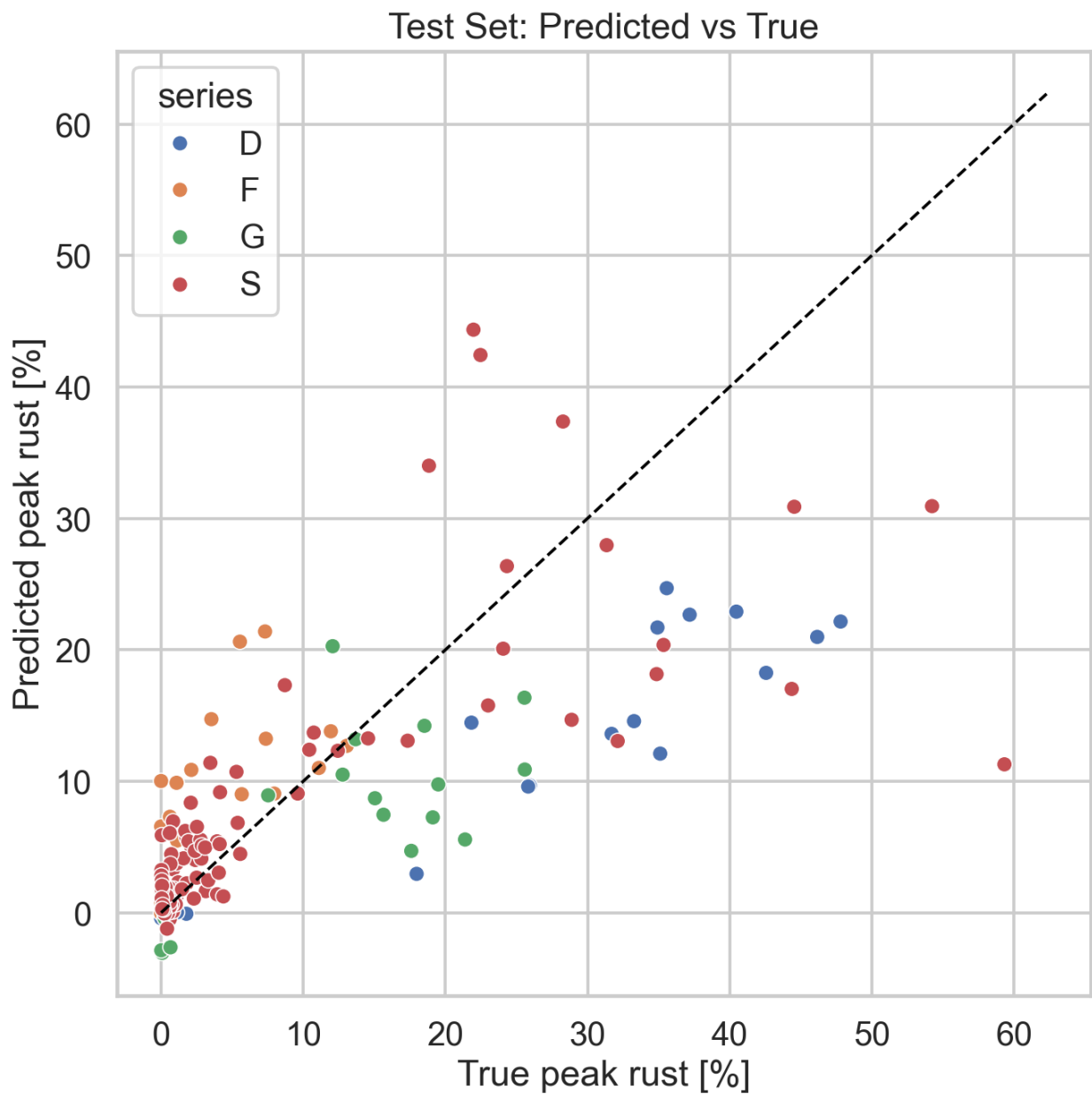


Figure 3. Residual distribution for error behavior analysis.

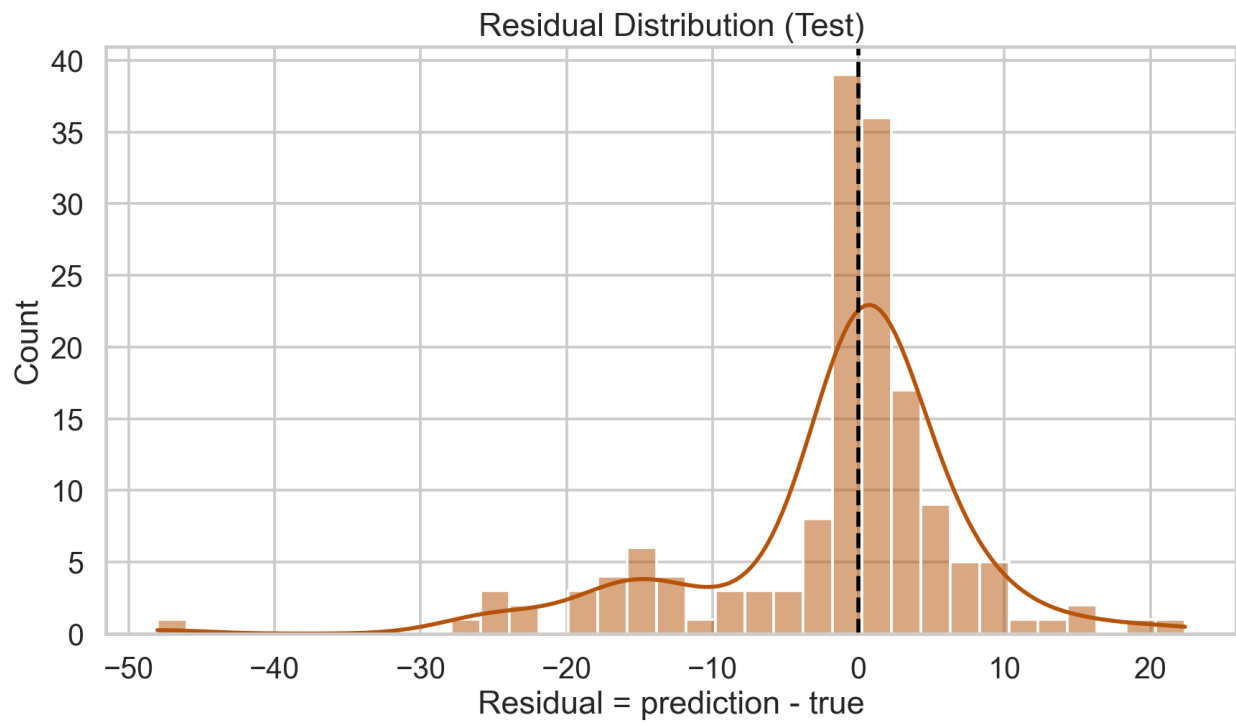


Figure 4. Progression patterns by treatment group.

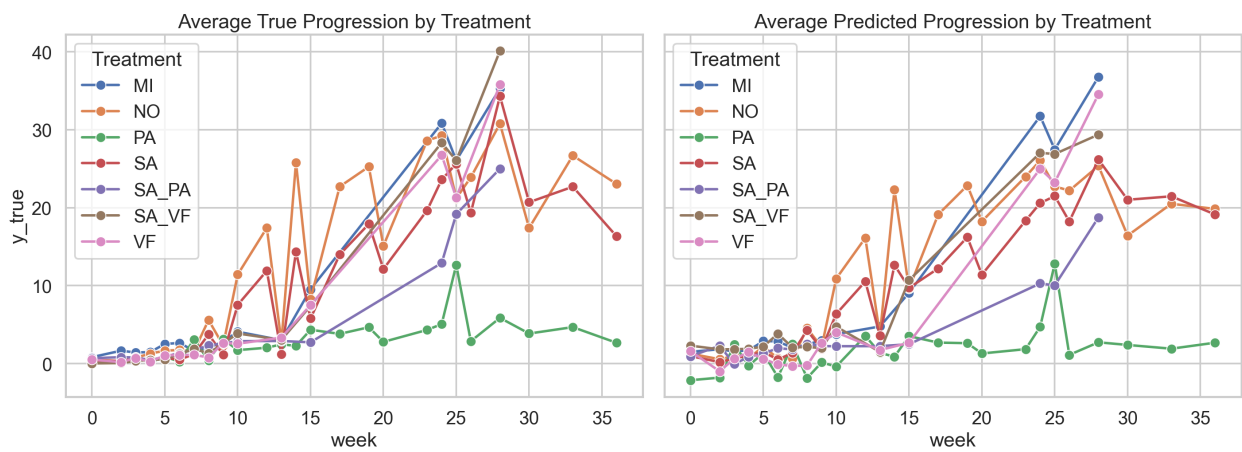
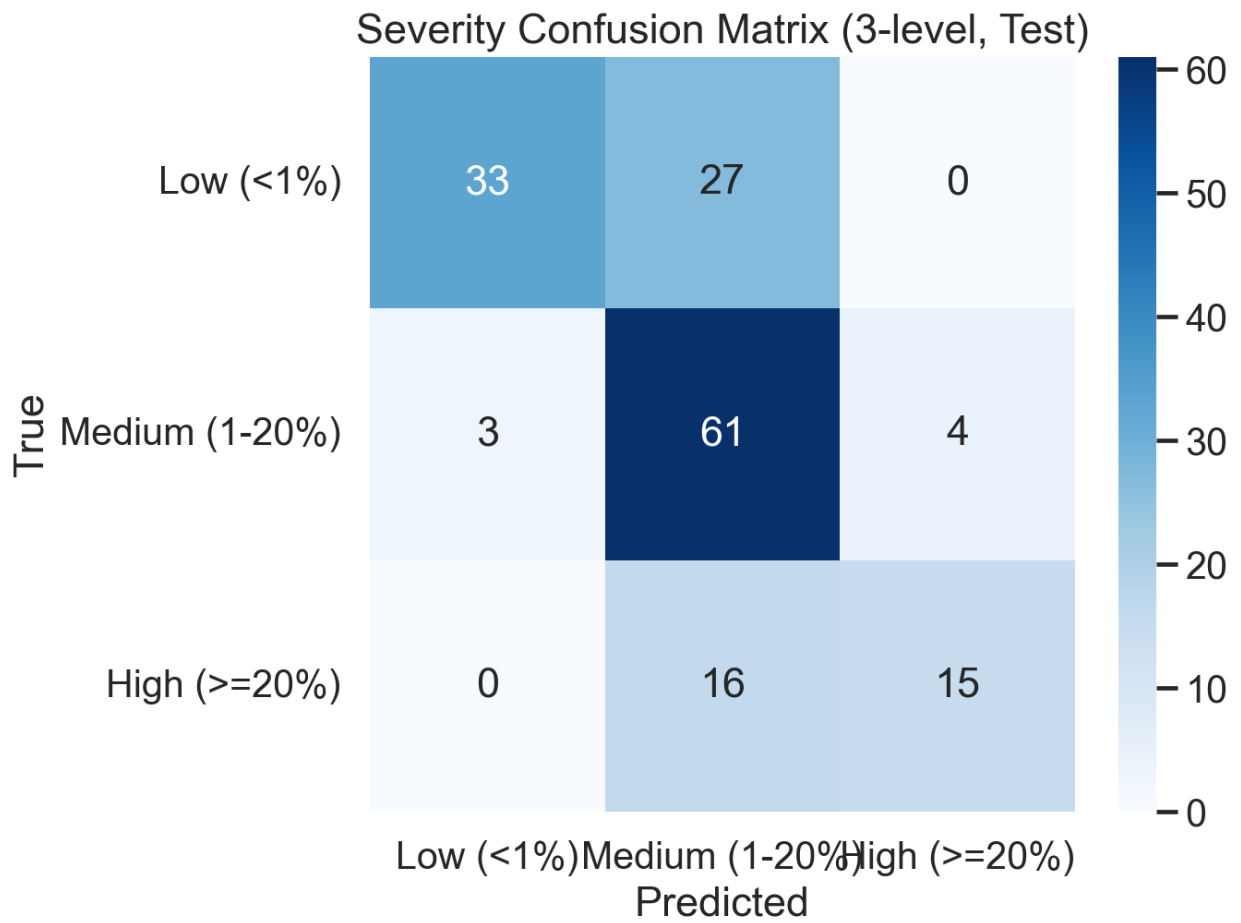


Figure 5. Three-level severity confusion matrix.



6. Full Material-Level Results

Table 5 reports per-material summaries across all predicted time points, including full material name and final-week prediction behavior.

	Treatment_Full_Name	n_obs	mae	mean_true	mean_prediction
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	1.2321	6.4726	6.6961
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	1.6368	8.7869	8.0232
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	0.7383	0.5164	0.4428
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	1.9093	46.9469	46.9039
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	13.8934	26.6223	12.8102
steel meshes: 4, NaCl: 5.0%)	No treatment (control)	18	5.6366	16.0182	17.1725
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	2.2572	10.5966	8.7723
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	3.9142	14.1578	14.2573
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	2.1305	14.4230	15.3972
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.1944	9.3267	8.7418
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.8836	13.1915	13.0819
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.6527	8.7854	9.2031
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	5.5848	4.3597	9.8911
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	4.2900	17.8439	14.6366
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.3406	9.7739	10.7293
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	13.6264	27.6836	14.1198
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.5210	10.5234	11.3055
corrosion inhibitor (steel meshes: 4, NaCl: 5.0%)	Surface-applied corrosion inhibitor	18	1.7756	16.2666	15.9590
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	1.2720	0.0010	-1.0050
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	6.4422	12.5159	7.2082
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	1.1121	0.0287	-0.8180
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	1.0908	0.0011	-0.3846
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	1.6294	0.3084	-0.7278
paint (steel meshes: 4, NaCl: 5.0%)	Painted protective treatment	18	1.7706	1.9080	0.6714
control (steel meshes: 7, NaCl: 3.5%)	No treatment (control)	15	1.3341	3.7373	4.6249
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	4.3914	4.8611	8.7464
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	2.6423	7.0901	8.8039
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	5.9839	12.4982	12.4621
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	1.7171	17.0681	16.2519
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	1.6754	6.5684	5.0242
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Mixed-in corrosion inhibitor	15	1.0577	2.2289	2.0360
control (steel meshes: 7, NaCl: 3.5%)	No treatment (control)	15	2.1738	8.2701	6.3902
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	0.8141	7.6718	7.6920
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	6.2454	12.6931	12.7748
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	2.2744	4.4311	6.1735
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	0.9440	6.4708	6.2172

	Treatment_Full_Name	n_obs	mae	mean_true	mean_pre
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	1.0352	5.4150	5.0057
corrosion inhibitor (steel meshes: 7, NaCl: 3.5%)	Surface-applied corrosion inhibitor	15	6.5510	9.7549	3.9260
treatment (steel meshes: 7, NaCl: 3.5%)	Painted protective treatment	15	1.5023	3.9989	3.6867
applied inhibitor (steel meshes: 7, NaCl: 3.5%)	Painting + surface-applied inhibitor	15	1.5481	3.6331	3.6768
applied inhibitor (steel meshes: 7, NaCl: 3.5%)	Painting + surface-applied inhibitor	15	4.0014	7.5041	4.5745
applied inhibitor (steel meshes: 7, NaCl: 3.5%)	Painting + surface-applied inhibitor	15	0.9881	4.2910	3.7175
control) (steel meshes: 7, NaCl: 3.5%)	No treatment (control)	15	2.5615	4.7566	6.0042
inhibitor + glass-based compound (steel meshes: 7, NaCl: 3.5%)	Surface-applied inhibitor + glass-based compound	15	3.6217	3.2797	5.0076
inhibitor + glass-based compound (steel meshes: 7, NaCl: 3.5%)	Surface-applied inhibitor + glass-based compound	15	2.2367	12.3472	11.0223
ol) (steel meshes: 7, NaCl: 3.5%)	No treatment (control)	15	1.9387	11.3275	10.9841
ound treatment (steel meshes: 7, NaCl: 3.5%)	Glass-based compound treatment	15	2.2559	9.5157	9.0305
ound treatment (steel meshes: 7, NaCl: 3.5%)	Glass-based compound treatment	15	1.5592	4.5499	3.8223

7. Interpretation, Alternatives, and Recommendations

- The image branch carried the dominant signal in this run, as the image-only model outperformed multimodal fusion on test MAE.
- Treatment-level differences indicate non-uniform prediction difficulty; this supports stratified data augmentation in future campaigns.
- High-error materials should be prioritized for additional data acquisition and repeated imaging quality checks.
- Phase 1 and Phase 2 should be used together in practice: Phase 1 for robust tabular forecasting and Phase 2 for richer visual quantification.

Possible solution tracks for the next project cycle:

- Track A: Fine-tune deeper CNN backbones (ResNet-50/EfficientNet) directly on corrosion regression.
- Track B: Build sequence-aware models using weekly image embeddings per specimen.
- Track C: Integrate uncertainty (deep ensembles or conformal prediction) for risk-aware maintenance decisions.
- Track D: Use survival/event models for predicting week-to-threshold with censored samples.
- Track E: Combine image model outputs with mechanistic corrosion priors from domain equations.

Practical suggestions:

- Standardize capture conditions even more tightly (illumination, distance, angle, white balance references).
- Collect more late-stage severe-corrosion examples to improve high-end extrapolation.
- Add quality-control flags in the dataset for blur/occlusion to prevent hidden label noise.
- Run periodic recalibration with new experiments and keep immutable train/validation/test splits for comparability.
- Deploy model monitoring to track drift by treatment group and specimen family.

Appendix material plots: 48 per-material progression figures are available in `/Users/youseffayyaz/Documents/Proj_corrosion/main_2/reports/figures/per_material` for detailed visual inspection.